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18/568,630	12/08/2023	Manuel SCHWARZ	080437.PG950US	8800

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EXAMINER	
SIDDIQUEE, ISMAAEEL ABDULLAH	

ART UNIT	PAPER NUMBER
3648	

NOTIFICATION DATE	DELIVERY MODE
11/26/2025	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/568,630	Applicant(s) SCHWARZ et al.	
	Examiner ISMAAEEL A SIDDIQUEE	Art Unit 3648	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 12/08/2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 11-20 is/are pending in the application.

5a) Of the above claim(s) 1-10 is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 11-20 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 12/08/2023 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Information Disclosure Statement

The information disclosure statement (IDS) submitted on **12/08/2023** is in compliance with the provisions of 37 CFR 1.97. Accordingly, the IDS is being considered by the examiner.

Examiner's Note

To help the reader, examiner notes in this detailed action claim language is in bold, ~~strikethrough~~ limitations are not explicitly taught and language added to explain a reference mapping are isolated from quotations via square brackets.

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

Claim(s) **11-12** is/are rejected under 35 U.S.C. 103 as being unpatentable over **Allouis et al. (US 20180294560 hereinafter Allouis)** in view of **Gourlay (US 20100296025)**.

Regarding claim **11,** Allouis teaches **An exterior component for a means of transportation** (Abstract "The invention relates to a radome"),

comprising: a light-guiding layer, which is transparent at least in sections (Abstract "tne light-emitting element which is coupled to the light guide such that light enters said light guide, propagates through the at least one light guide, is reflected on the reflective region, and enters the transparent front plate at said at least one light passage such that the light is emitted from the front face of the front plate."), ~~**having a depression**~~;

a light source disposed ~~**in the depression**~~ (fig 3),

which is configured to couple light into the light-guiding layer (Abstract "and at least one light-emitting element which is coupled to the light guide such that light enters said light guide");

a cover layer arrangement (0002 "Radomes are covering elements or covering plates that cover a radar transmitter or radar sensor which emits radar waves"), **which is transparent at least in sections, on the light-guiding layer** (0003 "In a radome, it is furthermore essential that it is at least substantially transparent for electromagnetic radiation in the longwave wavelength range"),

which is configured to at least partially reflect the light coupled into the light- guiding layer (0008 "An exemplary embodiment of the invention relates to a radome having a substantially extensive front plate, which is configured to be transmissive on the front side and which is provided on the rear side with a layer that does not transmit light, wherein at least one cutout in the form of a light passage is formed in the layer that does not transmit light, having a reflective region of the extensive front plate, wherein the reflective region is arranged next to the cutout"); **and**

a light decoupling device, which is configured to decouple at least a first part of the light out of the exterior component in an area of the exterior component spaced apart ~~from the depression~~, in which the exterior component is radar-transparent (0009 "According to the invention, it is advantageous if at least one diffuser element is arranged at the at least one light guide, which diffuser element is arranged such that the light propagating through the at least one light guide enters the diffuser element, propagates through the diffuser element, and exits the diffuser element and enters the transmissive front plate at the at least one light passage and is reflected at the reflective region such that it exits the front plate on the front side.").

Allouis does not explicitly teach the strikethrough limitations. However, in a related field of endeavor, Gourlay teaches

a light source disposed in the depression (0057 "At the perimeter interface between the substrate element (2) and the light guide layer (10), a cavity layer structure (not shown) may be incorporated in order to form a suitable cavity in which the LEDs may be embedded.")

Furthermore, it would have been obvious to one of ordinary skill in the art, at the time of filing of the instant application, to include the teachings of Gourlay with the teachings of Allouis. One would have been motivated to do so in order to advantageously improve light guide layers and structures (Gourlay 0005). Further still, the Supreme Court in KSR International Co. v. Teleflex Inc. (KSR), 550 U.S. 398, 82 USPQ2d 1385 (2007) provides that combining prior art elements according to known methods to yield predictable results may render a claimed invention obvious over such combination. Here, Gourlay merely teaches that it is well-known to incorporate the particular light guide features. Since both Gourlay and Allouis disclose similar light guides, one of ordinary skill in the art would recognize that the combination of elements here has previously been executed according to known methods, thereby evidencing that such combination would yield predictable results.

Regarding claim **12,** the cited prior art teaches **The exterior component according to claim 11, wherein the light-guiding layer has an index of refraction which is greater than 1.4 or greater than 1.45 or greater than 1.5, and/or the light-guiding layer is produced from polycarbonate** (Gourlay 0022 "The refractive index of the film layer is typically greater than 1.5."; 0023 "The substrate element of the film is typically made from polyester or polycarbonate sheet and the microstructures may be formed in an acrylic polymer")**.**

Furthermore, it would have been obvious to one of ordinary skill in the art, at the time of filing of the instant application, to include the teachings of Gourlay with the teachings of Allouis. One would have been motivated to do so in order to advantageously improve light guide layers and structures (Gourlay 0005). Further still, the Supreme Court in KSR International Co. v. Teleflex Inc. (KSR), 550 U.S. 398, 82 USPQ2d 1385 (2007) provides that combining prior art elements according to known methods to yield predictable results may render a claimed invention obvious over such combination. Here, Gourlay merely teaches that it is well-known to incorporate the particular light guide features. Since both Gourlay and Allouis disclose similar light guides, one of ordinary skill in the art would recognize that the combination of elements here has previously been executed according to known methods, thereby evidencing that such combination would yield predictable results.

Claim(s) **13-19** is/are rejected under 35 U.S.C. 103 as being unpatentable over **Allouis et al. (US 20180294560 hereinafter Allouis)** in view of **Gourlay (US 20100296025)** as applied to claim 11, and further in view of **Caruso et al. (US 20210384622 hereinafter Caruso)**.

Regarding claim **13,** the cited prior art teaches **The exterior component according to claim 12, ~~wherein the light source has a fiber-optic cable having at least one core and at least one casing, and the fiber-optic cable is embedded at least in sections in the light-guiding layer~~.**

The cited prior art does not explicitly teach the strikethrough limitations. However, in a related field of endeavor, Caruso teaches

wherein the light source has a fiber-optic cable having at least one core and at least one casing, and the fiber-optic cable is embedded at least in sections in the light-guiding layer (Caruso Claim 38 "optically connecting of the fibre optic device with at least one light engine via at least one connection element, wherein the connection element, preferably comprising a light guide and/or fibre optical strings, is it at least partly located in and/or connected to the intermediate layer, the sublayer, the film and/or the overmolded layer and/or optically connecting the substrate with the light engine.").

Furthermore, it would have been obvious to one of ordinary skill in the art, at the time of filing of the instant application, to include the teachings of Caruso with the cited prior art. One would have been motivated to do so in order to advantageously improve radome durability (Caruso 0077). Further still, the Supreme Court in KSR International Co. v. Teleflex Inc. (KSR), 550 U.S. 398, 82 USPQ2d 1385 (2007) provides that combining prior art elements according to known methods to yield predictable results may render a claimed invention obvious over such combination. Here, Caruso merely teaches that it is well-known to incorporate the particular light guide features. Since both the cited prior art and Caruso disclose similar light guides and housing structures, one of ordinary skill in the art would

recognize that the combination of elements here has previously been executed according to known methods, thereby evidencing that such combination would yield predictable results.

Regarding claim **14,** the cited prior art teaches **The exterior component according to claim 13, wherein the at least one casing is provided with a means for predefined decoupling of the light out of the fiber-optic cable into the light-guiding layer** (Caruso 0020 "Additionally, the direct overmolding of the decorative layer encases the layer, thereby protecting it from the elements, electrically isolating it, and reduces the likelihood of water ingress between the substrate and the overmolded layer, a problem encountered with multilayer radomes bound by adhesives.")**.**

Furthermore, it would have been obvious to one of ordinary skill in the art, at the time of filing of the instant application, to include the teachings of Caruso with the cited prior art. One would have been motivated to do so in order to advantageously improve radome durability (Caruso 0077). Further still, the Supreme Court in KSR International Co. v. Teleflex Inc. (KSR), 550 U.S. 398, 82 USPQ2d 1385 (2007) provides that combining prior art elements according to known methods to yield predictable results may render a claimed invention obvious over such combination. Here, Caruso merely teaches that it is well-known to incorporate the particular light guide features. Since both the cited prior art and Caruso disclose similar light guides and housing structures, one of ordinary skill in the art would recognize that the combination of elements here has previously been executed according to known methods, thereby evidencing that such combination would yield predictable results.

Regarding claim **15,** the cited prior art teaches **The exterior component according to claim 14, wherein the cover layer arrangement is materially bonded to the light-guiding layer** (Allouis 0022 "It is also expedient if the light guide and the diffuser element are arranged as interconnected parts in neighboring fashion, for example are interconnected in particular by way of injection molding or adhesive bonding")**.**

Regarding claim **16,** the cited prior art teaches **The exterior component according to claim 15, wherein the cover layer arrangement has a carrier layer having an optically visible pattern, and/or the cover layer arrangement has an intermediate layer** (Allouis fig 6 [shows a cover layer with an optically visible pattern])**, which is configured to reflect light incident on the intermediate layer from the light-guiding layer back in the direction of the light-guiding layer** (Allouis 0015 "It is also advantageous here if the reflective region is formed as a boundary face, as a lacquered boundary face or as a mirror-coated boundary face.")**.**

Regarding claim **17,** the cited prior art teaches **The exterior component according to claim 16, wherein the cover layer arrangement has a heating layer, which is radar-transparent at least in sections, and/or the cover layer arrangement has a surface coating, which is formed on a surface of the cover layer arrangement facing away from the light-guiding layer** (Caruso 0021 "the substrate and the decorative layer are heated prior to overmolding. Preferably, the substrate and the decorative layer are heated to at least 60 degrees Celsius, or to at least 70 degrees Celsius, or to at least 75 Degrees Celsius, or to at least 80 degrees Celsius prior to overmolding. This reduces the rate of temperature change in the decorative layer during the second-shot of the overmolding process, thereby decreasing the degree of thermal expansion during overmolding and helping to reduce the likelihood of visual distortion of the decorative layer.")**.**

Furthermore, it would have been obvious to one of ordinary skill in the art, at the time of filing of the instant application, to include the teachings of Caruso with the cited prior art. One would have been motivated to do so in order to advantageously improve radome durability (Caruso 0077). Further still, the Supreme Court in KSR International Co. v. Teleflex Inc. (KSR), 550 U.S. 398, 82 USPQ2d 1385 (2007) provides that combining prior art elements according to known methods to yield predictable results may render a claimed invention obvious over such combination. Here, Caruso merely teaches that it is well-known to incorporate the particular light guide features. Since both the cited prior art and Caruso disclose similar light guides and housing structures, one of ordinary skill in the art would recognize that the combination of elements here has previously been executed according to known methods, thereby evidencing that such combination would yield predictable results.

Regarding claim **18,** the cited prior art teaches **The exterior component according to claim 17, wherein the light decoupling device is formed at a point of the light-guiding layer opposite to the cover layer arrangement, and/or the cover layer arrangement is formed as a film** (Gourlay 0017 "mounting one or more light sources onto a first surface of a layer of film, wherein said layer of film has formed on a second surface a series of microstructures, wherein said microstructures allow light inside the layer of film to escape therefrom"; Abstract "This invention relates to film layer which is suitable for use in a light guide plate and methods of forming said film layer and light guide plate")**.**

Regarding claim **19,** the cited prior art teaches **A means of transportation having an exterior component according to claim 18, wherein the means of transportation is a vehicle, and/or the means of transportation has a radar system** (Allouis 0001 "The invention relates to a radome, in particular for a radar sensor for a motor vehicle.")**, which is configured to emit radar waves** (Allouis 0001 "a radar sensor")**, and the exterior component is arranged on the means of transportation such that the radar waves are incident on the radar-transparent area of the exterior component** (Caruso 0041 "It is thus desired to implement such additional functionalities in the radome without negatively influencing the core function to cover the radar antenna without influencing the transparency.")**.**

Furthermore, it would have been obvious to one of ordinary skill in the art, at the time of filing of the instant application, to include the teachings of Caruso with the cited prior art. One would have been motivated to do so in order to advantageously improve radome durability (Caruso 0077). Further still, the Supreme Court in KSR International Co. v. Teleflex Inc. (KSR), 550 U.S. 398, 82 USPQ2d 1385 (2007) provides that combining prior art elements according to known methods to yield predictable results may render a claimed invention obvious over such combination. Here, Caruso merely teaches that it is well-known to incorporate the particular light guide features. Since both the cited prior art and Caruso disclose similar light guides and housing structures, one of ordinary skill in the art would

recognize that the combination of elements here has previously been executed according to known methods, thereby evidencing that such combination would yield predictable results.

Claim(s) **20** is/are rejected under 35 U.S.C. 103 as being unpatentable over **Allouis et al. (US 20180294560 hereinafter Allouis)** in view of **Gourlay (US 20100296025),** and further in view of **Caruso et al. (US 20210384622 hereinafter Caruso)** as applied to claim 19, and further in view of **Kuwayama et al. (US 20110038178 hereinafter Kuwayama).**

Regarding claim **20,** the cited prior art teaches **A method for producing an exterior component for a means of transportation according to claim 18, comprising at least one injection-compression molding process by means of an injection-compression molding device** (Allouis 0022 "It is also expedient if the light guide and the diffuser element are arranged as interconnected parts in neighboring fashion, for example are interconnected in particular by way of injection molding or adhesive bonding"),

wherein the injection-compression molding process comprises the following steps: inserting (the cover layer arrangement, which is transparent at least in sections (Caruso 0030 "the functionality of the radar unit covered by the radome as the decorative layer is transparent for radar radiation and thus not negatively influencing a transmission of radar radiation."), **as an insert into an injection-compression molding tool of the injection-compression molding device** (Caruso Abstract "and overmolding at least the decorative layer with a radio-transmissive polymer to provide an overmolded layer as well as a decorative radome comprising a first layer comprising a, preferably radio-transmissive, polymer, the first layer having a front surface;"); **injecting a molding compound into a cavity of the injection-compression molding tool** (0093 "Subsequently, the method further includes (7) overmolding at least the decorative layer (24) with a radio-transmissive polymer to provide an overmolded layer

(26).”; fig 2) ~~**while the injection-compression molding tool is incompletely closed; closing a gate of the injection-compression molding device to prevent the molding compound from flowing back out of the cavity**~~**; and compressing the molding compound to bond the transparent light-guiding layer to the cover layer arrangement** (Caruso Claim 40 “wherein the embedding of the heating wire comprises at least partly melting of the intermediate layer, the sublayer and/or the film, ultrasonic, thermally, thermosonically and/or mechanically melting, using compression.”)**.**

The cited prior art does not explicitly teach the strikethrough limitations. However, in a related field of endeavor, Kuwayama teaches

while the injection-compression molding tool is incompletely closed; closing a gate of the injection-compression molding device to prevent the molding compound from flowing back out of the cavity (0011 “Therefore, for example, a remaining part formed when cutting off the gate section of the light guide plate manufactured by injection molding is not sandwiched between two unit light guide plates, and adjacent unit light guide plates are allowed to be arranged sufficiently closely, and light leakage between the unit light guide plates is allowed to be reduced. In other words, bright light generation is allowed to be inhibited.”).

Furthermore, it would have been obvious to one of ordinary skill in the art, at the time of filing of the instant application, to include the teachings of Kuwayama with the cited prior art. One would have been motivated to do so in order to advantageously improve the light source (Kuwayama 0029). Further still, the Supreme Court in KSR International Co. v. Teleflex Inc. (KSR), 550 U.S. 398, 82 USPQ2d 1385 (2007) provides that combining prior art elements according to known methods to yield predictable results may render a claimed invention obvious over such combination. Here, Kuwayama merely teaches that it is well-known to incorporate the particular light guide features. Since both the cited prior art and Kuwayama disclose similar light guides and housing structures, one of ordinary skill in

the art would recognize that the combination of elements here has previously been executed according to known methods, thereby evidencing that such combination would yield predictable results.

Conclusion

The prior art made of record and not relied upon is considered pertinent to application's disclosure:

a. **Ueno et al. (US 20050248844)** discloses "An optical path switching method according to the present invention comprises converging and irradiating, on a light absorption layer film provided in a thermal lens forming element (1, 2, 3) including at least the light absorption layer film, (See abstract)"

Any inquiry concerning this communication or earlier communications from the examiner should be directed to ISMAAEEL A. SIDDIQUEE whose telephone number is (571) 272-3896. The examiner can normally be reached on Monday-Friday 8am-5pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, William Kelleher can be reached on (571) 272-7753. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see https://ppair-my.uspto.gov/pair/PrivatePair. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a

USPTO Customer Service Representative or access to the automated information system, call 800-786-

9199 (IN USA OR CANADA) or 571-272-1000.

/ISMAAEEL A. SIDDIQUEE/
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